7
Nanocoatings for architectural glass

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Abstract: Window glass is an important building material that influences thermal and visual comfort in buildings. Glazings with spectrally selective coatings are suitable for many architectural applications. Thin film selective coatings can be used to modulate the optical and thermal properties of glass, and low emissivity glazings reduce radiative heat loss and aid passive solar control. Special glazings use chromogenic materials for dynamic modulation of light and solar transmittance in response to external stimuli such as temperature, solar radiation or electric current. A combination of chromogenic glazings with light redirecting systems, enhanced transparency and special protective and self-cleaning surfaces complete the list of smart glazing devices. This chapter presents an overview of several types of window glazings and coatings.

Key words: window glazings, glass coatings, low-emissivity, light transmittance, reflectance, solar radiation, daylighting, infrared radiation, wavelength.

7.1 Introduction

High-rise buildings with glazed curtain walls together with the drive to make them energy efficient have been largely responsible for development in the field of architectural glazings, and the main focus of window glass design has been on improving its optical and thermal properties.[1,2] Special highly transparent and heat-reflective glazings can be used to save energy by reducing heat loss through windows, as well as using solar transmittance control to minimise cooling and air conditioning costs.[1–3] Some window glazings are also used for light – enhancing and redirecting or decorative purposes.

Double or triple glass panes sealed together as one glass product with distant cavities are used as window glass units in temperate climate.[2,3] Radiative heat loss from a standard double-glazed, air filled unit is about double (about 65%) the heat loss through convection and conduction (about 35%).[4] Therefore, to achieve higher thermal performance of windows, radiative heat loss needs to be reduced.[2–4]

Thin film coatings deposited on the surface of a pane of glass during the manufacturing process modulate its optical and thermal properties.[1–4] The optical properties of thin films depend on the materials used and the layer composition in a coating as well as on the quality of the glass substrate surface and the coating deposition technology used.[5,6] There are two main groups of coatings used for architectural applications, as spectrally selective coatings for standard windows and as chromogenic thin films for switchable glass devices.

Glasses with spectrally selective coatings can be classified according to their influence on the indoor climate: as a glass with limited or enhanced light transmittance and as glazings with enhanced infrared reflectance.[7] Special angular selective thin films are also used for solar control applications.[2,8] The above-mentioned angular and spectrally selective glazings can be classified as passive, because their optical properties do not vary in response to external stimuli.

Chromogenic glazings can dynamically modulate solar radiation transmittance, and are used in intelligent switchable solar control systems. The glazing include chromogenic materials which facilitate reversible colouring procedures triggered by physical or chemical stimuli.[1,2,9–11] The chromogenic glazings can also be combined with solar holographic or microstructured transparent materials for smart glazing devices. Recently developed nanotechnology processes, focused on precise and premediated manipulation of atoms of materials, provide new possibilities for enhancing the physical properties, durability and self-cleaning effects of glazing systems.[12,13]

7.2 Spectrally selective glass

7.2.1 Optical properties and emissivity of glass

The optical properties of glass[5,6] are generally determined as spectral reflectance $\rho(\lambda)$, transmittance $\tau(\lambda)$ and absorptance $\alpha(\lambda)$ in a specified wavelength λ. These properties interrelate in the following way, based on the principle of energy balance:

$$\rho(\lambda) + \tau(\lambda) + \alpha(\lambda) = 1 \quad [7.1]$$

When designing architectural glass, it is very important to take into account the solar radiation affecting a glass pane in the spectral range $\lambda \in \langle 300\ \text{nm};2500\ \text{nm}\rangle$[14] and especially in the spectrum of visible light $\lambda \in \langle 380\ \text{nm};780\ \text{nm}\rangle$. Other important spectra include the near infrared solar range $\lambda \in \langle 780\ \text{nm};2500\ \text{nm}\rangle$ as well as the infrared (IR) radiation of building interior surfaces $\lambda \in \langle 3\ \mu\text{m};50\ \mu\text{m}\rangle$.[15]

Solar radiation transmitted through windows is partly absorbed by non-transparent interior surfaces. Solar radiation energy transforms into thermal energy once it has been absorbed. The building constructions are warmed and their surfaces emit IR radiation. Standard window glass is opaque for long-wavelength IR radiation[16] and there is also very low glass reflectance in the IR range, meaning that if $\rho(\lambda) \to 0$ and $\tau(\lambda) = 0$, absorptance is increased $\alpha(\lambda) \to 1$ in the long-wavelength IR range.

According to Wien's law,[17] if the temperature is 20 °C (room temperature) then the wavelength at which the radiation curve peaks is 10 μm. These temperature and wavelength values are important for the design of window glasses.

Stefan–Boltzmann's law[17] states that radiation between surfaces depends on emissivities (ε) of surfaces. Energy $M(\lambda,T)$ [Wm^{-2}] emitted from a real grey body at temperature T [K] is, according to Kirchhoff's law,[17] derived as:

$$M(\lambda,T) = \alpha(\lambda)\sigma T^4 = \varepsilon(\lambda)\sigma T^4 \qquad [7.2]$$

where $\alpha(\lambda)$ is absorptance, $\varepsilon(\lambda)$ is emissivity and σ is the Stefan–Boltzmann constant.[17]

It is clear from Eq. 7.2 that absorptance is directly proportional to emissivity in the IR range. As noted, standard window glass has very high IR absorptance. Practically, this means that standard window glass undergoes high heat radiation losses on cold days with low outdoor temperature, due to high glass emissivity.

On the basis of the aforementioned assumptions, the relationship between reflectance and emissivity in the IR range is:[18,19]

$$\rho(\lambda) = 1 - \varepsilon(\lambda) \qquad [7.3]$$

Heat loss through the glass is reduced due to a thin low emissivity coating deposited on its surface, preferably a coating with high visible transmittance for $\lambda \in \langle 380\text{ nm};780\text{ nm}\rangle$ and very high reflectance in spectral range $\lambda \in \langle 3\text{ μm};50\text{ μm}\rangle$. The higher the IR reflectance is, the lower the emissivity of the surface. Standard glass mean emissivity is $\varepsilon = 0.835$[3,19] and the mean IR reflectance reaches a value of $\rho = 0.165$. Low-emissivity glazings have ε between 0.1 and 0.05 or less, which indicates a high reflectance of 0.90 to 0.95 or even higher.[2,3,20]

7.2.2 Glass with low-emissivity coatings

The low-emissivity (low-e) glazings show spectral selectivity in high visible transmittance and high IR reflectance. Two types of coatings are used for the following practical applications:[3, 4,20–37]

- low-e coatings for reduction of heat losses through the glazing due to reflection of long-wavelength IR radiation back into the building interior;
- low-e solar control coatings for the elimination of solar gains due to the reflection of the near IR part of solar radiation into the exterior.

Doped oxide semiconductor thin films, consisting of dielectric layers with mobile charge carriers, can be used as low-e coatings. Materials such as SnO_2:F, In_2O_3:Sn, SnO_2:Sb, ZnO:Al or Cd_2SnO_4 are used.[20,22–27] Microgrid conducting coatings also prove low emissivity but they are not commonly used for window glazings.[20]

IR reflective films of dielectric/metal/dielectric multilayers are widely used where low-e coatings are required.[20–38] Metals such as silver, gold and copper have high IR reflectance[21,28,29] and silver is frequently used because it provides a high IR-reflective and colour neutral coating.[14,18,19] Gold-based reflective coating[36,37] technology is very expensive for window glazings and is not colour neutral. However, new nanomaterials offer gold nanoparticle coatings with optimised reflectance and colour.[38]

Thin dielectric films protect the metal film and increase its light transmittance. Frequently used dielectric materials for the low-e coatings are TiO_2, SnO_2, SiO_2, ZrO_2, ZnS, ZnO, $SnBO_2$, In_2O_3, Si_3N_4 and Bi_2O_3.[20–39] Dielectric layers serve an anti-reflective purpose in coatings. Protective layers such as seed and sacrificial layers are also applied in low-e coatings.[39] A seed layer undercoats the metal film and optimises crystalline growth of a thin layer of silver. A sub-oxidic interface layer on the silver film is also called the 'sacrificial layer'. An example of a typical low-e coating composition with anti-reflective, metal, seed and sacrificial layers can be found in Table 7.1.[39]

The aforementioned low-e coating exists in two variations as:

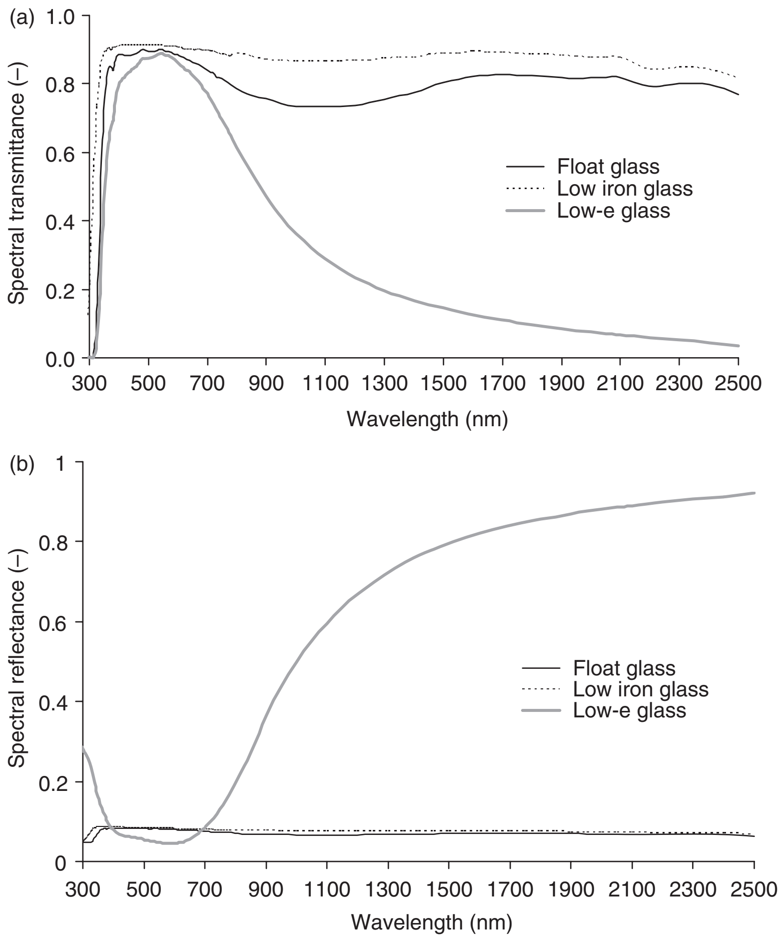
- *symmetrical coating*: glass/SiO_2/ZnO/Ag/TiO_x/SnO_2/SiN_xO_y with two anti-reflective layers of the same refractive indices;
- *asymmetrical coating*: glass/TiO_2/ZnO/Ag/TiO_x/SnO_2/SiN_xO_y with dielectric anti-reflective layers of different refractive indices: titanium dioxide with higher refraction index 2.5 and tin dioxide with lower refractive index 2.0 – this composition creates a low-e coating with higher light transmittance than the similar symmetrical coating.[39]

Table 7.1 Low-e coating layer

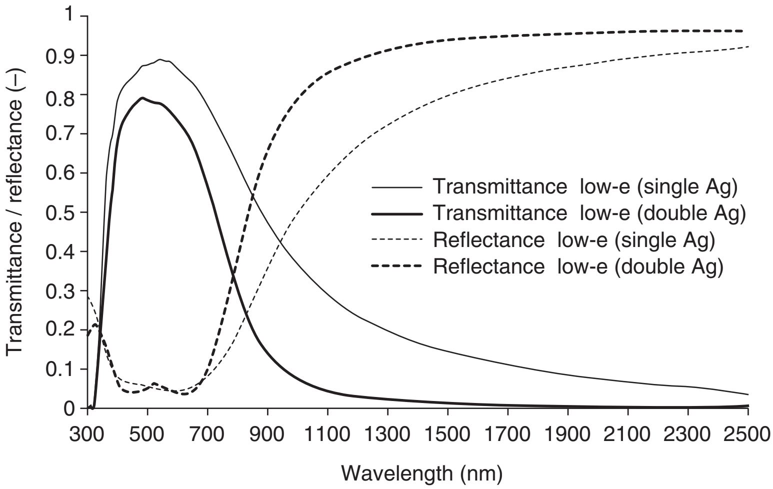
Layer	Material	Function of a layer in the low-e coating
Protective coating	SiN_xO_y	Chemically and mechanically resistant top layer
Anti-reflective layer	SnO_2	Protective and anti-reflection coating
Interface layer	TiO_x	Sub-oxidic layer
Metal layer	Ag	High infrared reflectance layer
Seed layer	ZnO	Optimises the silver layer growing to be very thin
Anti-reflective layer	SnO_2 (or TiO_2)	Increases light transmittance of the low-e coating
Glass substrate		

Low-e window coatings can be designed with double or even triple metal layer films.[39–42] These coatings also contain seed, anti-reflective and sacrificial layers but the silver film is separated into two or three thin layers that alternate with dielectrics. These types of multilayer dielectric/metal/dielectric coatings serve as broadband IR reflective coatings. The coatings reduce thermal radiation losses through the glass and also provide control due to reduction of solar IR transmittance.[40–42]

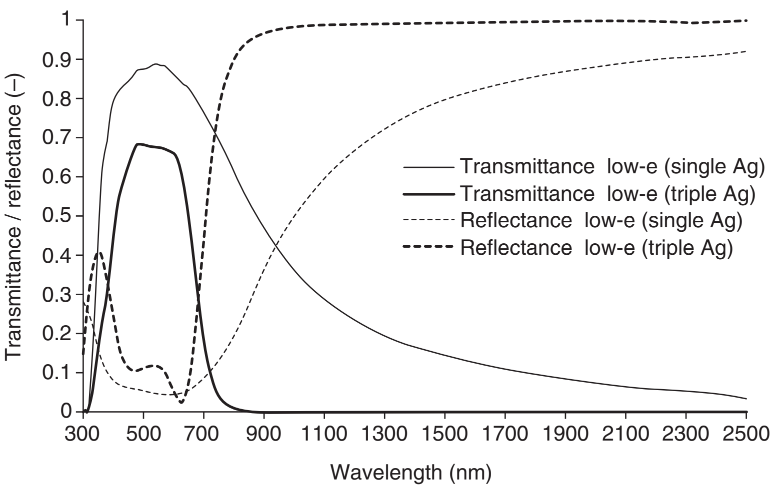
Spectral characteristics of a standard clear and a low-iron glass and a low-e glazing are compared in Fig. 7.1 for the solar radiation spectral range. Spectral transmittance and reflectance of double and triple silver low-e glazings compared to a single silver low-e glass are shown in Figs 7.2 and 7.3.



7.1 Spectral transmittance (a) and reflectance (b) of a clear float and low iron and low-e glass. (Saint Gobain).



7.2 Spectral transmittance and reflectance of a low-e glass with single and double Ag layer. (Saint Gobain).



7.3 Spectral transmittance and reflectance of a low-e glass with single and triple Ag layer. (Saint Gobain).

Spectrally selective multilayer films of polymers with different refractive indices represent another possibility for spectrally selective solar reflecting interference coatings. The coatings have colour neutrality and high light transmittance and IR reflectance in the spectral range between 780 and 2500 nm. They should be protected by a low-iron glass pane without any other coating on the exterior side.[43]

All of the previously mentioned spectrally selective coatings can be classified as passive functional coatings with invariable optical properties.

7.3 Dynamic smart glazings

Glazings with dynamic variations of solar transmittance depending on external stimuli are known as 'smart' glazings. They reflect or absorb the infrared part of solar radiation to minimise solar heat gains. Smart glazings use chromogenic materials to facilitate reversible darkening and bleaching,[44–49] and for this reason they are also called chromogenic glazings.

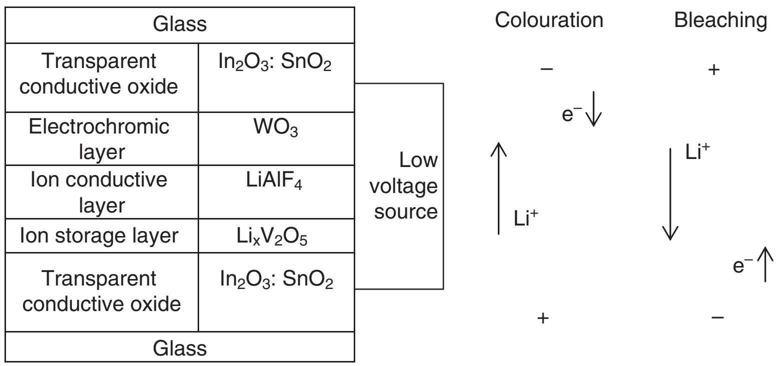
The dynamic variations in glass transmittance can be initiated by changes of temperature in thermochromic glazings or solar radiation in photochromic glass. Electrically activated chromogenic glazings are represented by a group of switchable devices such as electrochromic glazings and their possible modifications, as well as liquid crystal glazings and suspended particle devices. The electrically activated glazings use transparent conductive oxide coatings to establish electric fields when connected to a circuit.

7.3.1 Electrochromic glazings

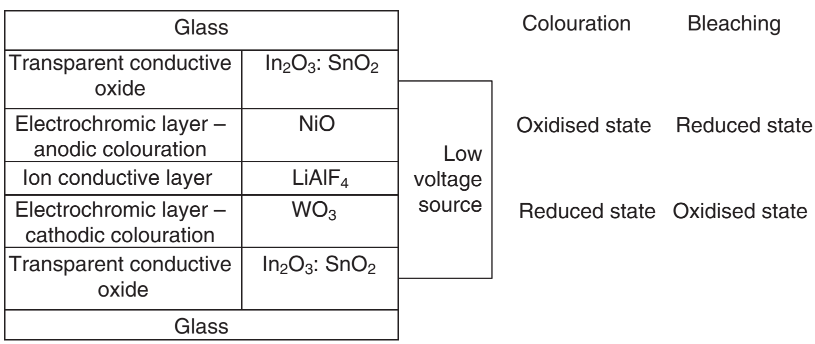
Electrochromic glazings are chromogenic devices that dynamically modify transmittance under voltage. The electrochromic devices are coloured due to an electrochemical reaction in the electrochromic thin films.[50–64] In the case of cathodic films, for example WO_3 or Nb_2O_5, the cause is electrochemical reduction; whereas in anodic films as IrO_2 or NiOOH or V_2O_5, the film becomes coloured due to electrochemical oxidation.[48–55]

The typical structure of an electrochromic glazing with a cathodic film is presented in Fig. 7.4.[61,62] The electrochromic layer (WO_3), ion conductor layer ($LiAlF_4$) and ion storage layer ($Li_xV_2O_5$) are sandwiched between two glass panes with transparent conductive oxides. The conductive layers are connected with a low voltage source. The applied voltage causes an electrochemical reaction within the ion conductor layer, or electrolyte. Ions from the ion storage layer are driven through the electrolyte into the electrochromic layer in the darkening state. Li^+ ions inserted into the electrochromic WO_3 layer induce absorption. The insertion (colouration) and extraction (bleaching) of the lithium ions is in balance with an equal and opposite movement of electrons (e^-) as the reaction progresses from the transparent state to the coloured state $WO_3 + ye^- + yLi^+ \leftrightarrow LiyWO_3$. Upon reversal of the applied potential, the ions migrate through the electrolyte into the ion storage layer and electrons are transported from the electrochromic layer via an external circuit.[60–62]

Some electrochromic glazing devices have the ion storage layer replaced by a complementary electrochromic layer.[62–64] In this type of glazing, one



7.4 Example of electrochromic glazing composition (cathodic colouration).



7.5 Example of glazing composition with two complementary electrochromic layers.

electrochromic layer causes cathodic colouration (coloured in the reduced state) and the complementary electrochromic layer exhibits anodic colouration (coloured in the oxidised state). The first electrochromic layer is reduced and the second oxidised by the electrical activation.

For example, a tungsten trioxide electrochromic layer is coloured in the reduced state and a nickel oxide layer is coloured in the oxidised state (see Fig. 7.5). The application of both electrochromic films supplies the desired level of optical modulation.[63,64] When the tungsten oxide layer is in the oxidised state, the nickel oxide layer is reduced, resulting in high transparency.

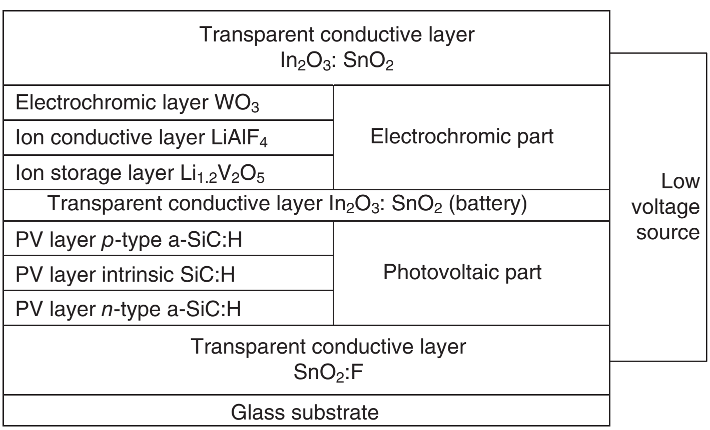
7.3.2 Photovoltaic-powered electrochromic glazings

Photovoltaic-powered electrochromic glazings combine both the photovoltaic and electrochromic thin films in one switchable device (see Fig. 7.6). The photovoltaic layers generate a voltage which causes colouration of the electrochromic film. A transparent, conductive layer between the photovoltaic and electrochromic layers can be used to charge an external battery. The top and bottom transparent conductive layers are used for circuit connection.[65–67]

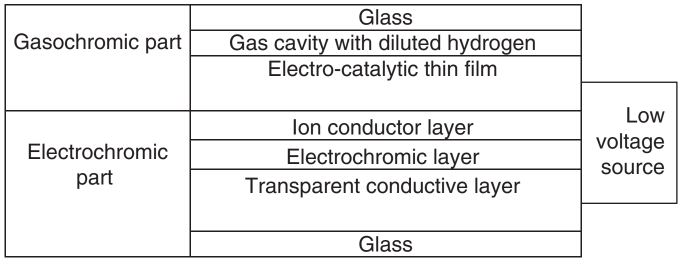
7.3.3 Gasochromic–electrochromic glazings

Colouration of gasochromic glazings is chemically activated.[61–64,68–70] A gasochromic device consists of an electrocatalytic thin coating based on tungsten trioxide deposited on glass. The glass is then sealed into a double glazed unit (the WO_3 layer with the catalytic film facing the inside of the unit cavity). The device transmittance variations occur when either diluted hydrogen or diluted oxygen, which activates the switching, is introduced into the cavity. A chemical reaction of hydrogen with the catalytic film on the WO_3 layer causes blue colouration. The reverse process (bleaching) is activated when the cavity is filled with diluted oxygen.[61]

Application of the gasochromic and electrochromic systems in one device represents an alternative to the electrically activated switchable glazing.[62–64,68–70] The double glazed unit has both clear (transparent conductive) and electrochromic (electrochromic and ion conductive) layers, with the electrocatalytic coating exposed to hydrogen, as seen in Fig. 7.7. The gas reacts with the catalyst that serves as the source of ions. The electrocatalytic



7.6 Photovoltaic-powered electrochromic device.



7.7 Composition of gasochromic electrochromic device.

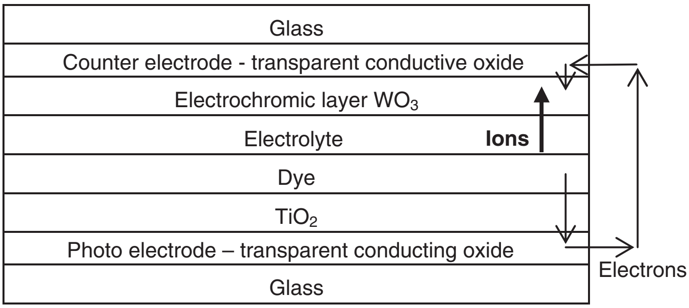
layer is connected with the transparent conductive layer and voltage is activated between them. Ions formed at the catalyst move through the ion conductive layer and are deposited into the electrochromic layer, which causes colouration. The bleaching process of the ion reversible transport is activated under reverse electric potential.

7.3.4 Photochromic-electrochromic glazing devices

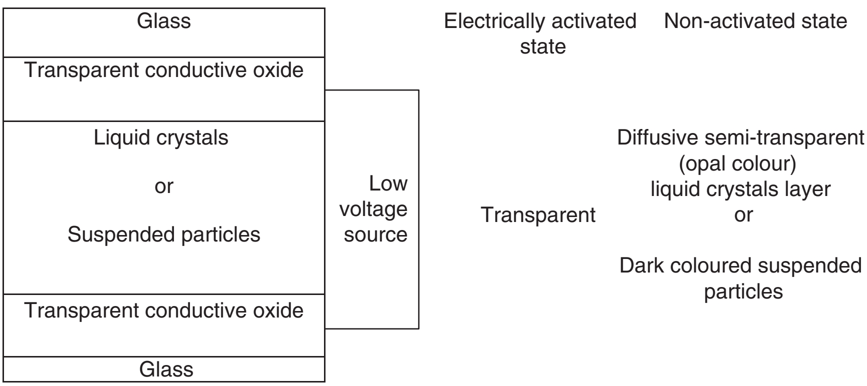
Photochromic glazings change their optical properties in response to radiation exposure, meaning that electrical activation is not required for colouration. Photosensitive materials are dispersed inside the glass or deposited as surface coatings.[71,72] Dispersed silver halide crystals in photochromic glass create colour centres which are activated under ultraviolet and short-wave visible radiation. Long-wave visible radiation triggers destruction of the colour centres so that the photochromic glass becomes transparent.

A photo-electrochromic glazing combines photochromic and electrochromic technology in one device.[73–75] A layer of electrolyte such as a lithium solid polymer containing lithium iodide is placed between a titanium dioxide and an electrochromic layer. The porous TiO_2 layer is dye-impregnated.[75] This three-layer thin film is sandwiched into two glass panes with transparent conducting oxide layers (see Fig. 7.8).

The glazing colouration is solar-activated: the dye absorbs solar radiation and releases electrons. The electrons are injected into the porous TiO_2 layer and move into the conducting oxide layer. They then pass through an external circuit to the conductive oxide layer adjacent to the electrochromic layer. The following flow of electrons through the central electrolyte towards the TiO_2 layer causes lithium ions to be injected into the electrochromic layer of WO_3 which causes colouration. Lithium ions are driven out of the electrochromic layer during the reversible bleaching process when solar rays do not affect the device.[75] Despite the non-electrical activation system, external voltage can be applied to the circuit for user control of the device.



7.8 Activated photoelectrochromic device.



7.9 Typical composition of a glass laminate with liquid crystals or suspended particles.

7.3.5 Liquid crystal and suspended particle glazings

Devices containing liquid crystals and suspended particles are similar to the glazing technologies already discussed, but the difference between them is light transmittance control. Liquid crystal glass is a light scattering device, while colouration of suspended particle glazing is due to light absorption (see Fig. 7.9).[76–84]

Liquid crystal glazing consists of a polymer matrix of microscopic cavities filled with liquid crystal molecules sandwiched between two glass panes, with transparent conductor layers connected to an external circuit.[77–79] Liquid crystal molecules align in planes perpendicular to the transparent conductor layers when an alternating electrical field is activated between the conductors. In its electrically activated state, the glazing is transparent. In the off-state, the liquid crystals' random configuration scatters light transmitted though the glazing.

The suspended particle device consists of two glass panes with transparent conductive layers and an internal layer of organic fluid with suspended particles. The suspended particles are inorganic opaque randomly configurated molecules that absorb light. They can align in the presence of an electric field activated between the devices' two transparent electrodes.[76–78]

Common liquid crystal and suspended particle glazing requires voltage in order to become transparent. A new self-switching liquid crystal glazing technology has been developed as a reverse mode, and works by doping the polymer liquid crystal matrix with photoconductive molecules.[83,84]

7.3.6 Thermally-activated glazings

The light transmittance of thermotropic and thermochromic glazing depends on variations in temperature. The two types of glazing use different methods of solar control: thermotropic glazing is a light scattering device, whereas thermochromic glass is reflective in its activated state.

Thermotropic glazing consists of two glass panes laminated with an internal gel layer. The gel is a mixture of two components with different refractive indices.[85–87] In the transparent state the components are homogenous. When temperature rises the components are separated and transmitted light is diffused.[87] Unlike liquid crystal technology, thermotropic glass does not require an external voltage source.

Solar control in thermochromic glass is achieved due to a material phase transformation in the glass coating. The transformation changes the material properties between the semiconductor and metallic phases.[88–94] Glass with a thermochromic coating is transparent in the semiconductor state but changes to become highly reflective in the metallic state. The reversible phase transformation is achieved using a doped vanadium dioxide VO_2 coating. Tungsten[92,93] or magnesium[94] serves as the dopant. Thermochromic and multilayer coatings of alternating TiO_2 and VO_2 layers[95] have been investigated for window coating applications.

7.3.7 Light control and thermal imaging glazings

Combination of chromogenic glazing technology with optical systems provides smart transparent devices for solar control or conversion and light imaging applications. This can be achieved using holographic thin films and fluorescent materials[96–98] as well as mirrors and prismatic surfaces that concentrate and redirect light.[99,100] New technologies also include light-emitting diodes (LEDs) embedded in glass laminates. Organic LEDs are used for large flexible transparent sheets.[101] Electrodes composed of carbon nanotubes and magnetic nanowires are used in glazed panels with electroluminescence effects.[102]

Anti-reflective coatings are used for switchable glazing to enhance light transmittance in the transparent state as well as for many solar applications and architectural purposes. Coatings of a low refractive index material with high porosity or with surface nanopatterns, as well as multilayer dielectric films alternating materials with high and low refractive indices, can be used to reduce glass reflectance.[103–111]

7.4 Glass surface protections

Special glass coating technologies can also serve a protective purpose, improving abrasion and stain resistance of the glass and facilitating disinfection and cleaning.[112–122]

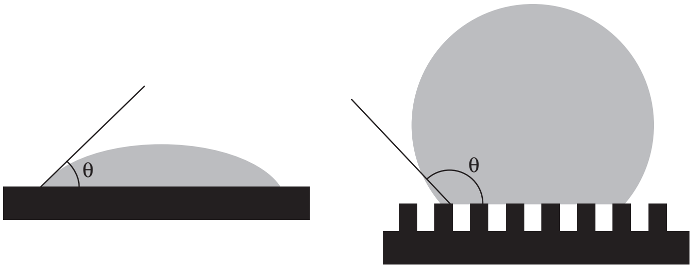
Ion beam milling is used to smooth the surface of an abrasion-resistant glass, and a diamond-like carbon coating is pyrolytically deposited on the surface. A similar technology can be used to create a stain-resistant surface. Surface sodium and sulphur concentrations are modified on the ion beam milled surface in an inert gas atmosphere. After this procedure a hydrophobic coating made up of a diamond-like carbon layer, and a fluoro-alkyl silane compound is deposited.[101,114]

Coatings containing dispersed silver ions serve an anti-bacterial function. The silver ion coating is convenient to use as a surface protection against bacteria or fungi.[115] Another coating of SiO_2 mixed with water or ethanol and known as 'liquid glass' can be applied to the surface to form an easy to clean and protective layer.[116]

Some coatings can reduce glass cleaning maintenance costs and prevent the glass from fogging. They can be classified as passive hydrophobic or hydrophilic coatings and active photocatalytic coatings that are activated due to solar radiation exposure.[117–122]

A glass hydrophobic coating contains silica nanoparticles embedded in an organic polymer matrix that creates a surface with nanoscale pores. The self-cleaning effect is evident.[119,120] Spherical drops are created on the hydrophobic surface due to water surface tension, meaning that the contact area of water droplets with the surface is minimised. The principle of artificial hydrophobic surfaces is based on natural observation (lotus plant leaves, grass, etc). The higher contact angle the higher the hydrophobicity of a surface. Surfaces with a contact angle less than $90°$ are considered to be hydrophilic and the other surfaces are referred to as hydrophobic (see Fig. 7.10).[119,120]

The glass self-cleaning effect based on a combination of hydrophilic and photocatalytic function is provided by a TiO_2 coating. The coating photocatalytically decomposes organic parts on its surface if activated under ultraviolet radiation. UV rays of solar radiation on the TiO_2 layer excite titanium and oxygen electrons migrate into the organic material on the



7.10 Water drop let contact angle on (a) hydrophilic surface $\theta < 90°$, (b) hydrophobic surface $\theta > 90°$.[119]

coating.[121,122] Rain water reacts with oxygen vacancies on the glass surface. The surface become hydrophilic and washes away the dirt.

A very special thin layer with carbon nanotubes can be applied as a protective layer to prevent condensation on glass. The layer connects to a source of electric current, creating a thin planar heating appliance that uniformly heats the glass area. The thin film has minimal heat capacity, meaning that heat is conducted into the glass substrate.[123]

7.5 Conclusion

Spectrally selective glazings for the reduction of heat radiation losses and solar infrared gains have wide architectural applications. Low emissivity glass coatings based on dielectrics with double or triple silver films provide light transmittance and more efficient reduction of the solar cooling load, compared to single silver low-e coatings.

Chromogenic glazings are dynamic solar control devices that respond to electrical, chemical, thermal or radiation triggers. Electrochromic glazings were developed through several design modifications and have been combined with integrated photovoltaics or gasochromic and photochromic systems. Thermochromic glazing is a convenient material for solar control in architectural structures. Liquid crystals and suspended particle devices are also used for interior glazing purposes.

Smart multifunctional windows, consisting of spectrally selective and switchable glazings with photovoltaics in combination with light-enhancing and solar concentration or redirection systems, can be used to create self-powered, glazed devices for energy efficiency and visual comfort in buildings. Improving the durability of smart window devices, as well as reducing production and maintenance costs, is an important task for future developments in architectural glazing.

7.6 Acknowledgements

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